NUMERICAL COMPUTING II

Assignment IV

Lagrange Interpolation

1. Given data: x = 0, 1, 2; f(x) = 1, 3, 7

1. Construct the Lagrange interpolating polynomial $P_2(x)$.

The Lagrange interpolation formula is $P_2(x) = \Sigma\, f(x_i)L_i(x)$, where $L_i(x) = \Pi\, (x - x_j)/(x_i - x_j)$, for $j \neq i$.

For $x_0 = 0$, $x_1 = 1$, $x_2 = 2$:

$L_0(x) = (x - 1)(x - 2) / [(0 - 1)(0 - 2)] = (x - 1)(x - 2)/2$

$L_1(x) = x(x - 2) / [(1 - 0)(1 - 2)] = -x(x - 2)$

$L_2(x) = x(x - 1) / [(2 - 0)(2 - 1)] = x(x - 1)/2$

Therefore, $P_2(x) = L_0(x) + 3L_1(x) + 7L_2(x)$.

Simplifying gives $P_2(x) = x^2 + x + 1$.

2. Write each Lagrange basis polynomial

$L_0(x) = (x - 1)(x - 2)/2$

$L_1(x) = -x(x - 2)$

$L_2(x) = x(x - 1)/2$

3. Verify the basis property

$L_i(x_j) = 1$ when $i = j$, and $L_i(x_j) = 0$ when $i \neq j$.

For example, $L_0(0) = 1$, $L_0(1) = 0$, $L_0(2) = 0$; $L_1(0) = 0$, $L_1(1) = 1$, $L_1(2) = 0$; $L_2(0) = 0$, $L_2(1) = 0$, $L_2(2) = 1$.

4. Estimate f(1.5)

$P_2(1.5) = (1.5)^2 + 1.5 + 1 = 2.25 + 1.5 + 1 = 4.75$.

5. Compare with the exact value

For $f(x) = x^2 + x + 1$, $f(1.5) = 4.75$. The interpolation result matches exactly.

Question 2: Interpolation from experimental data

Given (x, f(x)) = (1, 2.0), (2, 4.5), (3, 8.0), (4, 13.5).

The divided-difference coefficients are 2, 2.5, 0.5, and 0.5. The interpolating polynomial is $P_3(x) = 0.5x^3 - 0.5x^2 + 1.5x + 0.5$.

Estimate f(2.5): $P_3(2.5) = 6.75$.

Estimate f(3.5): $P_3(3.5) = 10.75$.

Comment: These estimates are between nearby measured values (4.5 and 8.0; 8.0 and 13.5, respectively), so they are reasonable for interpolation. The values should still be treated as estimates because the data are experimental.

Question 3: Non-uniformly spaced data

Given (x, f(x)) = (0, 1.000), (0.5, 1.125), (1.5, 1.875), (3, 4.000).

The data are not equally spaced. The Newton divided-difference coefficients are 1, 0.25, 0.25, and 0. Thus the interpolating polynomial simplifies to $P_3(x) = x^2/4 + 1$.

Estimate f(1): $P_3(1) = 1.25$.

Estimate f(2): $P_3(2) = 2.00$.

Lagrange interpolation is convenient for non-uniformly spaced data because it does not require equally spaced nodes.

Question 4: Given x = 2, 3, 5, 7; f(x) = 5, 10, 26, 50

The interpolating polynomial is $P_3(x) = (x^3 - 6x^2 + 17x - 6)/3$.

$P_3(4) = 17$, $P_3(6) = 37$, and $P_3(8) = 71$.

Of these estimates, f(6) is expected to be the most reliable because 6 lies between the given nodes 5 and 7. The values at 4 and 8 are also interpolations within the range of the data, but 4 is farther from the nearest node than 6; 8 is an extrapolation beyond the largest given node and is generally less reliable.

Question 5: Is every four-point interpolating polynomial cubic?

Given x = 1, 2, 3, 4 and f(x) = 1, 4, 9, 16.

The statement is not mathematically correct. Four distinct x-values determine a unique interpolating polynomial of degree at most 3, but its degree can be less than 3.

The data follow $f(x) = x^2$, so the interpolating polynomial is $P_2(x) = x^2$.

The resulting polynomial has degree 2, not 3.

In general, n distinct data points determine a unique polynomial of degree at most $n - 1$. The degree is lower when the data can be represented by a lower-degree polynomial.

Conclusion

Lagrange interpolation constructs a polynomial that passes through the given data points. It can be applied to equally spaced and non-uniformly spaced data. The examples show that the interpolating polynomial may have degree lower than the maximum possible degree. Estimates inside the range of the given data are generally more reliable than extrapolations outside that range.

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